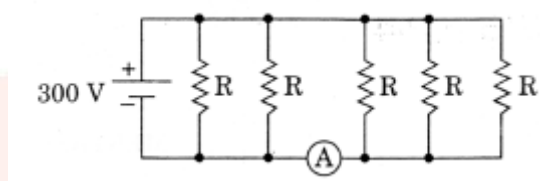


KCET – Physics – 2018
VERSION CODE: D

Question 1: Five identical resistors each of resistance $R = 1500 \, \Omega$ are connected to a 300 V battery as shown in the circuit. The reading of the ammeter A is



- (A) $\frac{1}{5} \text{ A}$
- (B) $\frac{3}{5} \text{ A}$
- (C) $\frac{2}{5} \text{ A}$
- (D) $\frac{4}{5} \text{ A}$

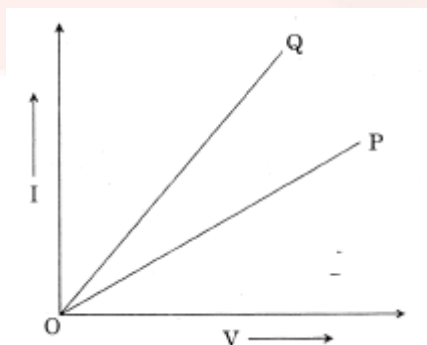
Answer 1: (B)

Question 2: Two cells of internal resistances r_1 and r_2 and of same emf are connected in series, across a resistor of resistance R . If the terminal potential difference across the cells of internal resistance r_1 is zero, then the value of R is

- (A) $R = 2(r_1 + r_2)$
- (B) $R = r_2 - r_1$
- (C) $R = r_1 - r_2$
- (D) $R = 2(r_1 - r_2)$

Answer 2: (C)

Question 3: The $I - V$ graphs for two different electrical appliances P and Q are shown in the diagram. If R_P and R_Q be the resistances of the devices, then



(A) $R_p = R_Q$

(B) $R_p > R_Q$

(C) $R_p < R_Q$

(D) $R_p = \frac{R_Q}{2}$

Answer 3: (B)

Question 4: The correct Biot-Savart law in vector form is

A) $d\vec{B} = \frac{\mu_0}{4\pi} \frac{I(d\vec{l} \times \vec{r})}{r^2}$

(B) $d\vec{B} = \frac{\mu_0}{4\pi} \frac{I(d\vec{l} \times \vec{r})}{r^3}$

(C) $d\vec{B} = \frac{\mu_0}{4\pi} \frac{Id\vec{l}}{r^2}$

(D) $d\vec{B} = \frac{\mu_0}{4\pi} \frac{Id\vec{l}}{r^3}$

Answer 4: (B)

Question 5: An electron is moving in a circle of radius r in a uniform magnetic field B .

Suddenly the field is reduced to $\frac{B}{2}$. The radius of the circular path now becomes

(A) $\frac{r}{2}$

(B) $2r$

(C) $\frac{r}{4}$

(D) $4r$

Answer 5: (B)

Question 6: A charge q is accelerated through a potential difference V . It is then passed normally through a uniform magnetic field, where it moves in a circle of radius r . The potential difference required to move it in a circle of radius $2r$ is

(A) $2V$

(B) $4V$

(C) $1V$

(D) $3V$

Answer 6: (B)

Question 7: A cyclotron's oscillator frequency is 10 MHz and the operating magnetic field is 0.66 T. If the radius of its dees is 60 cm, then the kinetic energy of the proton beam produced by the accelerator is

- (A) 9 MeV
- (B) 10 MeV
- (C) 7 MeV
- (D) 11 MeV

Answer 7: (C)

Question 8: Needles N_1 , N_2 and N_3 are made of a ferromagnetic, a paramagnetic and a diamagnetic substance respectively. A magnet when brought close to them will

- (A) Attract all three of them
- (B) Attract N_1 strongly, N_2 weakly and repel N_3 weakly
- (C) Attract N_1 strongly but repel and N_3 weakly
- (D) Attract N_1 and N_2 strongly but N_2 repel N_3

Answer 8: (B)

Question 9: The strength of the Earth's magnetic field is

- (A) Constant everywhere
- (B) Zero everywhere
- (C) Having very high value
- (D) Varying from place to place on the Earth's surface 10×10^{-3} s

Answer 9: (D)

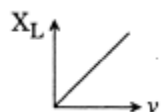
Question 10: A jet plane having a wing-span of 25 m is travelling horizontally towards east with a speed of 3600 km/hour. If the Earth's magnetic field at the location is $4 \times 10^{-4} T$ and the angle of dip is 30° , then, the potential difference between the ends of the wing is

- (A) 4V
- (B) 5V
- (C) 2V
- (D) 2.5V

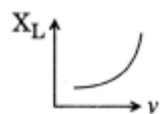
Answer 10: (B)

Question 11: Which of the following, represents the variation of inductive reactance (X_L) with the frequency of voltage source (ν)?

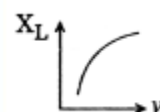
(A)



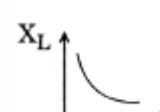
(B)



(C)



(D)



Answer 11: (A)

Question 12: The magnetic flux linked with a coil varies as $\phi = 3t^2 + 4t + 9$. The magnitude of the emf induced at $t = 2$ seconds is

- (A) 8V
- (B) 16V
- (C) 32V
- (D) 64V

Answer 12: (B)

Question 13: A 100 W bulb is connected to an AC source of 220 V, 50 Hz. Then the current flowing through the bulb is

- (A) $\frac{5}{11}$ A
- (B) $\frac{1}{2}$ A
- (C) 2A
- (D) $\frac{3}{4}$ A

Answer 13: (A)

Question 14: In the series LCR circuit, the power dissipation is through

- (A) R
- (B) L
- (C) C
- (D) Both L and C

Answer 14: (A)

Question 15: In Karnataka, the normal domestic power supply AC is 220 V. 50 Hz. Here 220 V and 50 Hz refer to

- (A) Peak value of voltage and frequency
- (B) rms value of voltage and frequency
- (C) Mean value of voltage and frequency
- (D) Peak value of voltage and angular frequency

Answer 15: (B)

Question 16: A step-up transformer operates on a 230 V line and a load current of 2 A. The ratio of primary and secondary windings is 1: 25. Then the current in the primary is

- (A) 25A
- (B) 50A
- (C) 15A
- (D) 12.5A

Answer 16: (B)

Question 17: The number of photons falling per second on a completely darkened plate to produce a force of 6.62×10^{-5} N is 'n'. If the wavelength of the light falling is 5×10^{-7} m, then $n = \underline{\hspace{1cm}} \times 10^{22}$.

- (A) 1
- (B) 5
- (C) 0.2
- (D) 3.3

Answer 17: (B)

Question 18: An object is placed at the principal focus of a convex mirror. The image will be at

- (A) Centre of curvature
- (B) Principal focus
- (C) Infinity
- (D) No image will be formed

Answer 18: (D)

Question 19: An object is placed at a distance of 20 cm from the pole of a concave mirror of focal length 10 cm. The distance of the image formed is

- (A) + 20 cm
- (B) + 10 cm
- (C) -20 cm
- (D) -10 cm

Answer 19: (C)

Question 20: A candle placed 25 cm from a lens forms an image on a screen placed 75 cm on the other side of the lens. The focal length and type of the lens should be

- (A) + 18.75 cm and convex lens

- (B) – 18.75 cm and concave lens
- (C) + 20.25 cm and convex lens
- (D) – 20.25 cm and concave lens

Answer 20: (A)

Question 21: A plane wavefront of wavelength λ is incident on a slit of width a . The angular width of principal maximum is

- (A) $\frac{\lambda}{a}$
- (B) $\frac{2\lambda}{a}$
- (C) $\frac{a}{\lambda}$
- (D) $\frac{a}{2\lambda}$

Answer 21: (B)

Question 22: In a Fraunhofer diffraction at a single slit, if yellow light illuminating the slit is replaced by blue light, then diffraction bands

- (A) Remain unchanged
- (B) Become wider
- (C) Disappear
- (D) Become narrower

Answer 22: (D)

Question 23: In Young's double slit experiment, two wavelengths $\lambda_1 = 780$ nm and $\lambda_2 = 520$ nm are used to obtain interference fringes. If the n^{th} bright band due to λ_1 coincides with $(n + 1)^{\text{th}}$ bright band due to λ_2 then the value of n is

- (A) 4
- (B) 3
- (C) 2
- (D) 6

Answer 23: (C)

Question 24: In Young's double slit experiment, slits are separated by 2 mm and the screen is placed at a distance of 1.2 m from the slits. Light consisting of two wavelengths 6500 Å and 5200 Å are used to obtain interference fringes. Then the separation between the fourth bright fringes of two different patterns produced by the two wavelengths is

- (A) 0.132mm
- (B) 0.123mm
- (C) 0.213mm
- (D) 0.412mm

Answer 24: (A)

Question 25: The maximum kinetic energy of emitted photoelectrons depends on

- (A) Intensity of incident radiation
- (B) Frequency of incident radiation
- (C) Speed of incident radiation
- (D) Number of photons in the incident radiation

Answer 25: (B)

Question 26: A proton and a α particle are accelerated through the same potential difference V . The ratio of their de Broglie wavelengths is

- (A) $\sqrt{2}$
- (B) $2\sqrt{2}$
- (C) $\sqrt{3}$
- (D) $2\sqrt{3}$

Answer 26: (B)

Question 27: The total energy of an electron revolving in the second orbit of hydrogen atom is

- (A) -13.6 eV
- (B) -1.51 eV
- (C) -3.4 eV
- (D) Zero

Answer 27: (C)

Question 28: The period of revolution of an electron in the ground state of hydrogen atom is T . The period of revolution of the electron in the first excited state is

- (A) $2T$
- (B) $4T$
- (C) $6T$
- (D) $8T$

Answer 28: (D)

Question 29: The energy equivalent to a substance of mass 1 g is

- (A) $18 \times 10^{13} J$
- (B) $9 \times 10^{13} J$
- (C) $18 \times 10^6 J$
- (D) $9 \times 10^6 J$

Answer 29: (B)

Question 30: The half-life of tritium is 12.5 years. What mass of tritium of initial mass 64 mg will remain undecayed after 50 years?

- (A) 32 mg
- (B) 8 mg
- (C) 16 mg
- (D) 4 mg

Answer 30: (D)

Question 31: In a CE amplifier, the input ac signal to be amplified is applied across

- (A) Forward biased emitter-base junction
- (B) Reverse biased collector-base junction
- (C) Reverse biased emitter-base junction
- (D) Forward biased collector-base junction

Answer 31: (A)

Question 32: If $A = 1$ and $B = 0$, then in terms of Boolean algebra, $A + \bar{B} =$

- (A) B
- (B) $\bar{B}$
- (C) A
- (D) $\bar{A}$

Answer 32: (C)

Question 33: The density of an electron-hole pair in a pure germanium is $3 \times 10^{16} \text{ m}^{-3}$ at room temperature. On doping with aluminium, the hole density increases to $4.5 \times 10^{22} \text{ m}^{-3}$. Now the electron density (in m^{-3}) in doped germanium will be

- (A) 1×10^{10}
- (B) 2×10^{10}
- (C) 0.5×10^{10}
- (D) 4×10^{10}

Answer 33: (B)

Question 34: The de common emitter current gain of a n-p-n transistor is 50. The potential difference applied across the collector and emitter of a transistor used in CE configuration is, $V_{CE} = 2 \text{ V}$. If the collector resistance, $R_C = 4 \text{ k}\Omega$, the base current (I_B) and the collector current (I_C) are

- (A) $I_B = 10 \mu\text{A}, I_C = 0.5 \text{ mA}$
- (B) $I_B = 0.5 \mu\text{A}, I_C = 10 \text{ mA}$
- (C) $I_B = 5 \mu\text{A}, I_C = 1 \text{ mA}$
- (D) $I_B = 1 \mu\text{A}, I_C = 0.5 \text{ mA}$

Answer 34: (D)

Question 35: The radius of the Earth is 6400 km. If the height of an antenna is 500 m, then its range is

- (A) 800 km
- (B) 100 km
- (C) 80 km
- (D) 10 km

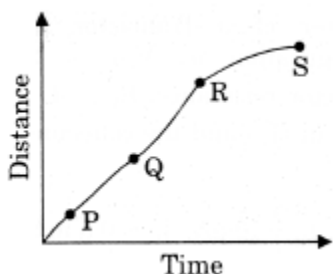
Answer 35: (C)

Question 36: A space station is at a height equal to the radius of the Earth. If ' v_E ' is the escape velocity on the surface of the Earth, the same on the space station is _____ times v_E .

- (A) $\frac{1}{2}$
- (B) $\frac{1}{4}$
- (C) $\frac{1}{\sqrt{2}}$
- (D) $\frac{1}{\sqrt{3}}$

Answer 36: (C)

Question 37: A particle shows distance-time curve as shown in the figure. The maximum instantaneous velocity of the particle is around the point

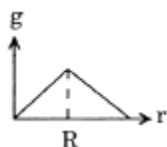


- (A) P
- (B) S
- (C) R
- (D) Q

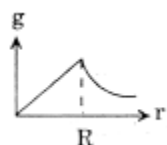
Answer 37: (C)

Question 38: Which of the following graphs correctly represents the variation of ' g ' on the Earth?

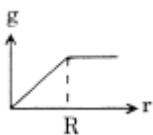
- (A)



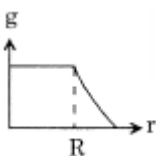
- (B)



- (C)



(D)



Answer 38: (B)

Question 39: A cup of tea cools from 65.5°C to 62.5°C in 1 minute in a room at 22.5°C . How long will it take to cool from 46.5°C to 40.5°C in the same room?

- (A) 4 minutes
- (B) 2 minutes
- (C) 1 minutes
- (D) 3 minutes

Answer 39: (A)

Question 40: The dimensions of the ratio of magnetic flux (ϕ) and permeability (μ) are

- (A) $[M^0 L^1 T^0 A^1]$
- (B) $[M^0 L^{-3} T^0 A^1]$
- (C) $[M^0 L^1 T^1 A^{-1}]$
- (D) $[M^0 L^2 T^0 A^1]$

Answer 40: (A)

Question 41: A mass 'm' on the surface of the Earth is shifted to a target equal to the radius of the Earth. If 'R' is the radius and 'M' is the mass of the Earth, then work done in this process is

- (A) $\frac{mgR}{2}$
- (B) mgR
- (C) mgR
- (D) $\frac{mgR}{4}$

Answer 41: (A)

Question 42: First overtone frequency of a closed pipe of length ' l_1 ' is equal to the 2nd harmonic frequency of an open pipe of length ' l_2 '. The ratio $\frac{l_1}{l_2} =$

(A) $\frac{3}{4}$

(B) $\frac{4}{3}$

(C) $\frac{3}{2}$

(D) $\frac{2}{3}$

Answer 42: (A)

Question 43: The resistance $R = \frac{V}{I}$ where $V = (100 \pm 5)V$ and $I = (10 \pm 0.2)A$. The percentage error in R is

(A) 5.2%

(B) 4.8%

(C) 7%

(D) 3%

Answer 43: (C)

Question 44: A block rests on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of static friction between the block and the plane is 0.8. If the frictional force on the block is 10 N, the mass of the block is ($g = 10 \text{ ms}^{-2}$)

(A) 1kg

(B) 2kg

(C) 3kg

(D) 4kg

Answer 44 : (B)

Question 45: Two particles of masses m_1 and m_2 have equal kinetic energies. The ratio of their momenta is

(A) $m_1 : m_2$

(B) $m_2 : m_1$

(C) $\sqrt{m_1} : \sqrt{m_2}$

(D) $m_1^2 : m_2^2$

Answer 45: (C)

Question 46: The pressure at the bottom of a liquid tank is **not** proportional to the

- (A) Acceleration due to gravity
- (B) Density of the liquid
- (C) Height of the liquid
- (D) Area of the liquid surface

Answer 46: (D)

Question 47: A Carnot engine takes 300 calories of heat from a source at 500 K and rejects 150 calories of heat to the sink. The temperature of the sink is

- (A) 125 K
- (B) 250 K
- (C) 750 K
- (D) 1000 K

Answer 47: (B)

Question 48: Pressure of an ideal gas is increased by keeping temperature constant. The kinetic energy of molecules

- (A) Decreases
- (B) Increases
- (C) Remains same
- (D) Increases or decreases depending on the nature of gas

Answer 48: (C)

Question 49: A man weighing 60 kg is in a lift moving down with an acceleration of 1.8 ms^{-2} . The force exerted by the floor on him is

- (A) 588 N
- (B) 480 N
- (C) Zero
- (D) 696 N

Answer 49: (B)

Question 50: Moment of inertia of a body about two perpendicular axes X and Y in the plane of lamina are 20 kg m^2 and 25 kg m^2 respectively. Its moment of inertia about an axis perpendicular to the plane of the lamina and passing through the point of intersection of X and Y axes is

- (A) 5 kg m^2
- (B) 45 kg m^2
- (C) 12.5 kg m^2
- (D) 500 kg m^2

Answer 50: (B)

Question 51: Two wires A and B are stretched by the same load. If the area of cross-section of wire 'A' is double that of 'B', then the stress on 'B' is

- (A) Equal to that on A
- (B) Twice that on A
- (C) Half that on A
- (D) Four times that on A

Answer 51: (B)

Question 52: The magnitude of point charge due to which the electric field 30 cm away has the magnitude 2 NC^{-1} will be

- (A) $2 \times 10^{-11} \text{ C}$
- (B) $3 \times 10^{-11} \text{ C}$
- (C) $5 \times 10^{-11} \text{ C}$
- (D) $9 \times 10^{-11} \text{ C}$

Answer 52: (A)

Question 53: A mass of 1 kg carrying a charge of 2 C is accelerated through a potential of 1 V. The velocity acquired by it is

- (A) $\sqrt{2} \text{ ms}^{-1}$
- (B) 2 ms^{-1}
- (C) $\frac{1}{\sqrt{2}} \text{ ms}^{-1}$
- (D) $\frac{1}{2} \text{ ms}^{-1}$

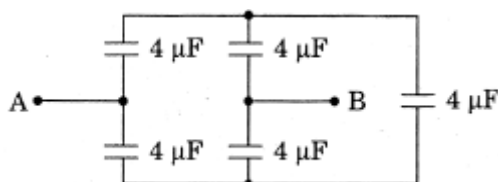
Answer 53: (B)

Question 54: The force of repulsion between two identical positive charges when kept with a separation 'r' in air is 'F'. Half the gap between the two charges is filled by a dielectric slab of dielectric constant = 4. Then the new force of repulsion between those two charges becomes

- (A) $\frac{F}{3}$
- (B) $\frac{F}{2}$
- (C) $\frac{F}{4}$
- (D) $\frac{4F}{9}$

Answer 54: (D)

Question 55: For the arrangement of capacitors as shown in the circuit, the effective capacitance between the points A and B is (capacitance of each capacitor is $4\mu F$)



- (A) $4\mu F$
- (B) $2\mu F$
- (C) $1\mu F$
- (D) $8\mu F$

Answer 55: (A)

Question 56: The work done to move a charge on an equipotential surface is

- (A) Infinity
- (B) Less than 1
- (C) Greater than 1
- (D) Zero

Answer 56: (B)

Question 57: Two capacitors of $3\mu F$ and $6\mu F$ are connected in series and a potential difference of 900 V is applied across the combination. They are then disconnected and reconnected in parallel. The potential difference across the combination is

- (A) Zero
- (B) 100 V
- (C) 200 V
- (D) 400 V

Answer 57: (D)

Question 58: Ohm's Law is applicable to

- (A) Diode
- (B) Transistor
- (C) Electrolyte
- (D) Conductor

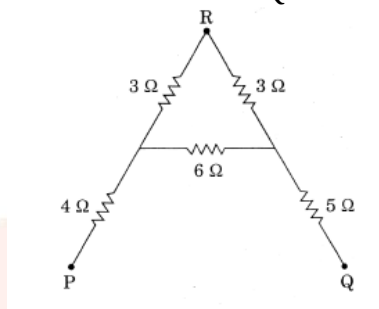
Answer 58: (D)

Question 59: If the last band on the carbon resistor is absent, then the tolerance is

- (A) 5%
- (B) 20%
- (C) 10%
- (D) 15%

Answer 59: (B)

Question 60: The effective resistance between P and Q for the following network is



- (A) $\frac{1}{12} \Omega$
- (B) 21Ω
- (C) 12Ω
- (D) $\frac{1}{21} \Omega$

Answer 60: (C)